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Avinash Gorde,
Fl.No15, Atharva Complex,
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Hello team,

I did my M.Tech.in Mathematical Modeling and Simulation from Centre for Modeling and Simulation,University of Pune. During M.Tech. I have also applied myself to learn modeling subjects like Differential Equation, Optimization, Numerical Linear Algebra, Numerical Methods, Machine Learning, Probability for Modelers, Stochastic Simulation, Stochastic Optimization and Evolutionary computing.

With over 9.6 years of experience in software development and a strong focus on machine learning and high-performance computing on the C++ and Python platform. I have a proven track record of designing and implementing cutting-edge scalable solutions. I have consistently tackled complex challenges by combining a deep understanding of software engineering principles with expertise in numerical methods, statistical modeling, and parallel computing. My academic foundation in Mathematical Modeling Simulation (M.Tech.) further demonstrates my ability to design robust algorithms and scalable architectures.

Working in WeAgile Software, ML and AI engineer on C++, CUDA, Python, GPU platform. Developing model inference using deep learning, for end to end video analytics pipeline. Deploying with Kubernetes and docker

Worked as a software engineer in Boeing, Siemens Digital Industry, Geometric-HCL

Sincerely yours,

Avinash Valiba Gorde,

Avinash Valiba Gorde

Email-id: avgorde@gmail.com (+91) 8999446209

OBJECTIVE

Learning new things puts the ability of analytical and logical skills to its best use for the benefit of the organization and me

WORK EXPERIENCE 9.6 YEARS, C++11.X, PYTHON, AI

Senior Software Engineer Machine Learning Aug-23- Apr-25 1.9 years
WeAgile Software solution, Pune Full-time

- Client-StrataVision Inc(previous DeepNorth), USA
- Building enterprise level **HPC**, scalable Computer Vision(CV) based cognitive pipelines. Developing cutting edge perception algorithms- **Detection, Tracking, Classification, Segmentation** using the NVIDIA tech stack, **DeepStream** using ML and AI for the retail industry on the GPU platform.
- Programming on **C++17, Python, CUDA, cuDNN, PyTorch, TensorRT, ONNX Runtime, Google PubSub and MPI** libraries for **Deep Learning**, Video analytics, video processing, Real Time Streaming Protocol (**RTSP**), DeepSORT tracker.
- Kernel development for detection models(YOLOX) using **CUDA**, Parallel computing.
- MLOps, DevOps, CI/CD github actions, docker image, package creation, Model deployment to partner cloud platform using **Kubernetes, Docker**, GCP and Google PubSub. ZMQ asynchronous messaging protocol.
- POC for **GenAI**, LLM-based(LLaMA-3) chatbot using LangChain, integrated with FastAPI, fine tune with LoRA, used vectordb **FAISS**, MCP, capable of context retention, querying and dynamic tool invocation **RAG**.
- Mathematical model inference using the the High Performance Inference framework, training of models tried various **optimizer** Adams, Adagrad, Gradient Descent. GPU profiling, bench marking on **Linux** platform, **A/B testing and validation**.

SDE-2(Model Dev.) Oct21- Aug23 1.10 years
Boeing, Bangalore Full-time

- **Mathematical Model** development using **C++ 11/14/17**, Monte carlo simulation, Simulation framework, PDE model, Stochastic Optimization, Discrete Event Simulation using **HPC, CUDA**. **Quantitative statistical development on multi threaded low latency using Quantlib, KDB, Q and high throughput framework, asynchronous messaging protocol**.
- Plugin Development for main application using **C++, STL, Python**
- Development of ML algorithm GA using TSP for Surveillance of aircraft and path optimization.
- Experimented with AutoGen and Crew AI to prototype multi-agent collaborative AI systems for task planning and automated reasoning.
- Platform and vehicle detection/classification using Unreal Engine, Airsim(High Fidelity Simulations) on **GPU** platform. Training model, specifying appropriate loss functions, optimizes, and hyper parameters. Data structure and algorithms, low level design(LLD), OOAD.
- **Porting of Python Numpy, Skyfield(Satellite Positioning library)** into C++. Working with a global team of Australia and the United Kingdom

Software Engineer(Product Dev.) Dec18-Oct21

2.10 years

Siemens PLM Software, Pune Full-time

- **Server-side development** using C++ and client side using **JAVA** in **PLM** reporting and analytics projects. Worked on a predictive modeling application that gives predictions based on historical data of the month, seasons, and which component will fail the most in the season. Teamcenter is enterprise PLM software for managing product data. Worked on dashboard creation to display analytical reports, line, column, pie charts.
- ML technique classification used grouping of item into the respective group
- **SOA, Client-Server, MCP based architecture**, dump data into database using **PLM ITK**.
- Built an information retrieval system combining Elasticsearch with custom NLP pipelines in spaCy and NLTK, improving search relevance for domain-specific queries. Design Pattern **MVC**
- C++, core Java, Javascript, plm database, vectordb, Python, R, ITK

Software Engineer Jul16 - Dec18

2.6 year

Geometric HCL, Pune, Full-time

- Object oriented design, Visual Studio, Qt, C++, STL, Boost library, sharedPtr, weakPtr, dump domain data using SQLite, XML, JSON, OpenScenegraph, visualization, qml, qml component, knowledge of Bitbucket atlassian community, Git-client(source tree), svn, Quickbuild.
- Knowledge of Design Pattern **Singleton, Visitor, Observer**.
- **Worked on Outlier Removal Algorithm, Principal Component Analysis(PCA), Statistical Modeling, Predictive Modeling in orthodontics based cad development project.**
- Numerical method based bisection algorithm finding culprit in data, least square fit for fitting polynomial curves, predicting best orthodontic setup from various patients data. Statistical graphs kurtosis based decision etc.
- Unit test and perform unit testing using google test(g-test) framework. Worked in agile software development.
- Participate in product related R&D discussion with US client.
- Python Jupyter Note Book, scikit learn, pandas.

Research Engineer(Product Dev.) Oct15-Jul16

8 months

Fluidyn Software Pvt. Ltd. Bangalore, Full-time

Pre-processing development using **Qt-C++** in Fluidyn-Multiphysics, Object oriented programming, Data structure & algorithm, STL, boundary conditions, Performs impact assessment, critical production issues and contributes to Root Cause Analysis of issues and outlines corrective actions Bug Fixing.

Project Intern Dec14-Jun2015***National Param Super computing Facility, Pune, Full-time***

Completed M.Tech project Conjugate heat transfer for electronic cooling using OpenFOAM.

ACADEMIC PROJECTS**M.Tech. Thesis**

2014-15

Conjugate heat transfer for electronic cooling using OpenFOAM

Understand the structure of C++ based OpenFOAM computational fluid dynamics library.
Develop mathematical simulation model, use parallel algorithm for faster simulation, validate experimental results with computational results.

B.E. Project

2011-2012

Design of solar concentrator with automatic tracking system

EDUCATION

M.Tech.(Mathematical Modeling & Simulation) 2013-2015
University of Pune

- College : Center for Modeling and Simulation, University of Pune.
- GPA : 4/6

B.E.(Mechanical Engineering) 2008-2013
University of Pune

- College : S.V.I.T, Nashik.
- Percentage : 63.86%

SKILLS

- Programming Languages **C, C++-11X, Python, KDB/Q, CUDA, PyTorch**
- Programming with various Packages **Matlab , R.**, OS- Linux, MAC iOS, Windows
- High Performance Computing(**HPC**), Parallel computing, using **CUDA, MPI**.
- Statistical Modeling, Regression, Machine Learning, Deep Learning
- Linear Algebra, Numerical Methods, Optimization, Partial Differential Equations.
- Certifications ”**Course in Machine Learning**” from **IIT-Madras**.

VOLUNTEER ACTIVITIES

I was volunteer at Siemens technology conclave.

PERSONAL DETAILS

<i>Email</i>	avgorde@gmail.com
<i>Birth</i>	June 25, 1988
<i>Gender</i>	Male
<i>Passport</i>	Z5860605
<i>Address</i>	Fl.No15, Atharva Complex, Bhosale Village, Fursungi Pune 412308
<i>Mobile</i>	91-8999446209, 91-9764242663
<i>Languages</i>	English, Marathi, Hindi

DECLARATION

I hereby declare that above mentioned-information is correct up to my knowledge and I bear the responsibility for the correctness of the above mentioned particular.

Avinash Gorde